and understanding, are known only to him who has himself experienced them."

Remarkably, Einstein was not the first to discover the correct form of the law of warpage, the form that obeys his relativity principle. Recognition for the first discovery must go to Hilbert. In autumn 1915, even as Einstein was struggling toward the right law, making mathematical mistake after mistake, Hilbert was mulling over the things he had learned from Einstein's summer visit to Göttingen. While he was on an autumn vacation on the island of Rugen in the Baltic the key idea came to him, and within a few weeks he had the right law—derived not by the arduous trial-and-error path of Einstein, but by an elegant, succinct mathematical route. Hilbert presented his derivation and the resulting law at a meeting of the Royal Academy of Sciences in Göttingen on 20 November 1915, just five days before Einstein's presentation of the same law at the Prussian Academy meeting in Berlin.

Quite naturally, and in accord with Hilbert's own view of things, the resulting law of warpage was quickly given the name the *Einstein field equation* (Box 2.6) rather than being named after Hilbert. Hilbert had carried out the last few mathematical steps to its discovery independently and almost simultaneously with Einstein, but Einstein was responsible for essentially everything that preceded those steps: the recognition that tidal gravity must be the same thing as a warpage of spacetime, the vision that the law of warpage must obey the relativity principle, and the first 90 percent of that law, the Einstein field equation. In fact, without Einstein the general relativistic laws of gravity might not have been discovered until several decades later.

Box 2.6

The Einstein Field Equation: Einstein's Law of Spacetime Warpage